Preparing for the In-Class Midterm

> ⚠ Under Construction
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> You are reading an **early draft** of the Practice Midterm! I want to make it available for eager beavers who are particularly worried about what the Midterm might look like, but I will announce over Canvas (and remove this message) when the final version is ready for you to work through.

The **in-class midterm** will be held on the following days, based on your section:

- **Section 01:** Monday, November 2, 2026
- **Section 02:** Thursday, November 5, 2026

The purpose of the midterm is twofold:

- First, to assess your comprehension of the course topics from Weeks 1 through 8, and their interrelationships, as they relate to the goals of modern data-driven organizations
- Second, to perform this assessment not in the abstract but via the development of a **concrete plan** that will then serve as a key step in your Final Project!

Your first step, in preparing for the midterm, is to **read through the dataset vignettes** and decide which one is most interesting to you. Once you have decided, **register your decision on Canvas** by choosing an option within the "Midterm Dataset Choice" assignment.

Finally, you can think through how the following "pieces" of what we've learned up to this point in the course would work relative to your chosen dataset!

Question 1: OLTP Infrastructure

This question is **started for you** in the vignette for your chosen dataset, where (in the subsection with header "OLTP Structure") you can get a sense for the **existing OLTP infrastructure** at the organization you've been hired at. In each case, your job is to think through **transforming** the information in this OLTP setup into the kinds of insights/inferences described at the beginning of the vignette.

Question 2: Ingesting, Transforming, and Loading OLTP Snapshots

Question 3: OLAP Infrastructure

Sample Question (Enron Email Corpus)

As you can see from the example emails in the vignette, both the content of the emails themselves and their metadata vary wildly in terms of what might be streaming into the OLTP DB (which is just recording the raw text+metadata for every incoming and outgoing email): for example, sometimes the content of the email is included as simple, human-readable plaintext, but sometimes the email body essentially just "points to" a set of attached files. Given this variability, and given how frequently emails are streaming in at this 10,000+ employee organization, which of the following conclusions would you draw for a "best-fit" OLAP infrastructure? *(Choose one)*

(a) A Data Lake infrastructure where all raw data from each email is "dumped" into an S3 bucket (for downstream schema-on-read parsing) makes most sense, since the bandwidth required to write the emails into an S3 bucket would be significantly lower than the bandwidth required to write data from the emails into a warehouse DB table.
(b) A Data Lake infrastructure where all raw data from each email is "dumped" into an S3 bucket (for downstream schema-on-read parsing) makes most sense, since it's difficult to plan in advance what specific "pieces" of the emails might be relevant.
(c) A Data Warehouse infrastructure where email data is written into a table with a pre-defined schema makes most sense, since we can determine in advance what data we want to record and pre-allocate columns within the warehouse DB accordingly
(d) A Data Warehouse infrastructure where email data is written into a table with a pre-defined schema makes most sense, since we can just allocate a "misc_data" column (of raw binary type) to hold any data from the emails that may not fit within the pre-defined schema

Here the answer would be (b), since it's the better choice of infrastructure paired with the better justification for using that infrastructure. (a) is "close" to the right answer, but wouldn't be marked as correct because it's an irrelevant justification (writing into a data warehouse DB doesn't require any more or less bandwidth than writing into a data lake S3 bucket, in general!)

Question 4: Querying OLAP Data

Question 5: Scaling OLAP Queries

Question 6: Serving and Reporting Our Scaled OLAP Query Results